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# Quantum classifier for recognition and identification of leaf profile features

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**Abstract.** Quantum-based classifiers and architecture are gaining lots of attention in image representation and cryptography. The proposed algorithm applies a quantum classifier to a computer vision system for leaf recognition which can be applied to a quantum computer. Images from ten species of leaves which are categorised into two groups, namely simple and palmately, are recognised using a quantum classifier. The pixels of images are transformed to qubit states using quantum Fourier transform (QFT) and Hadamard gates. The profile and structural features are extracted by applying 1D-convolution and controlled not (CNOT) gates. A quantum nearest neighbour search classifier is used to find the closest matching leaf based on probability. The results for different levels of image processing are evaluated and compared with the nearest neighbour classifier. The recognition rate of the quantum classifier for the best level of image processing is 97.33%. The recognition rate of the classifier is better than the nearest neighbour classifier and also has a low computation time.

## 1 Introduction

There are around 298,000 species of plants in the world of which 82,356 are uncategorized [29]. With the increasing risk of extinction of plants, it is essential to set up a database of plants for its protection [40]. An image database of leaves can be used and processed for information on plants. Image processing involves the signal processing of an input image. The source of the image can be a still image camera or a video. The output from image processing can be an image with less noise and more information or it can be image features. The majority of image processing techniques engross transformation techniques like edge detection for data reduction. The output of the image is dependent on the methodology of capturing the image.

The leaf of a plant contributes vastly to the taxonomy of a plant. Plant leaf can be classified by observing some of its morphological features like leaf vein structure, curls and nature of edges (smooth or hard), shape, and stem and colour [30]. Mainly the taxonomy data of leaves are present in the structure of the leaf or retrieved from databases online from other research works [12]., [30]. Leaves, in reality, have a complex three-dimensional structure, but images are of two dimensions; therefore, classification using images becomes difficult [6, 32]. The diverse appearances and complex structure of plants make leaf recognition a challenging task in a computer vision system (Sulc and Matas, [38] and for agriculture experts. Deep feature learning techniques like convolutional neural networks (CNN) make it easy to recognize complex structures; however, a quantum classifier makes the recognition fast and easy.

In this paper, we use a quantum circuit and transforms to retrieve the profile of leaves and use a quantum classifier to recognize 10 different types of leaves. The features of leaves are extracted by a quantum version of 1D-CNN and by hand-crafted geometrical features. The recognition is achieved using the quantum nearest neighbour circuit.

The literature survey on leaf classification and the use of quantum principles in computer vision is discussed in related works. An overview of image processing and image acquisition is presented in Sect. 3. Description of the features and the quantum-based feature extraction is discussed in Sect. 4. Section 5 gives an overview of the classification task and also compares the results of the proposed method with existing methods. The paper

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concludes with a discussion on the performance of the classifier and identification of leaf profiles.

## 2 Related works

Recognizing leaves and plants is very important in agriculture [39] and nutraceutical studies. Leaf recognition can be achieved in many ways. The recognition methodology for leaves has evolved in the past decades. Leaf classification using probabilistic neural network (PNN) [13, 40] and shape retrieval techniques [30, 32] has been overpowered by convolutional neural networks (CNN) (Fatihah et al., [14]. Some techniques which have been applied for leaf recognition are principal component analysis (PCA) followed by a PNN [40], learning vector quantization (LVQ) with radial bias function (RBF) [32], combined support vector machine with PCA, nearest neighbour with locally linear embedding (LLE) [6], artificial neural network (ANN) [12] and CNN [17], Fatihah et al., [14].

Many fields in computer vision and pattern recognition have been inspired by quantum systems. Tasks such as storing and retrieving images [1, 24], colour conversion [24], removing noise from images (Abura'ed, [2], Chakraborty, [11], Jankowski, [19], image representation (Abdulmunem, [1], Abura'ed, [2], Cai, [8], Chakraborty, [11, 18], Jankowski, [19, 23], Schuld, [34], Yuan, [41], Zhang, [42] and processing [18], Jankowski, [19], Zhang, [42] can be realized using the principals of quantum circuits and systems. The rows and the columns of image pixels can be represented as qubit pairs [18], Zhang, [42]. The quantum entanglement principals when applied to computer vision have shown great results in weighting and classification for bimodal computer vision system [23].

Quantum classifiers are used to achieve many tasks which include Higgs analysis [5], noise reduction (Grant, [16], Schuld [35], and to enhance machine learning techniques [7]. A binary quantum classifier has useful applications like feature selection from hyperspectral images [31]. The bases of a quantum classifier are quantum bits or qubits and quantum gates. Quantum gates can be used to implement different classifiers from distance-based to machine learning classifiers [9]. Mostly, the classical classifiers are extended to the quantum version for fast and robust output. Some quantum versions of classifiers are quantum competing networks [43], quantum neural networks (Gandhi, [15, 25], Sang, [33], and quantum nearest neighbour.

## 3 Proposed work

The main objective of this work is to recognize leaves using a quantum classifier. The main steps involved are image processing, qubit representation of images, and extracting the features from the qubit representation of images. Quantum gates are used to extract features and classify leaves. The performance is visualised using the Quirk simulator, which is available online.

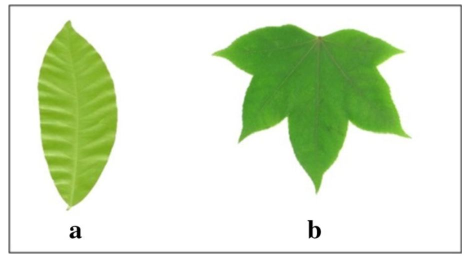


**Fig. 1** Leaf types, **a** simple **b** palmately

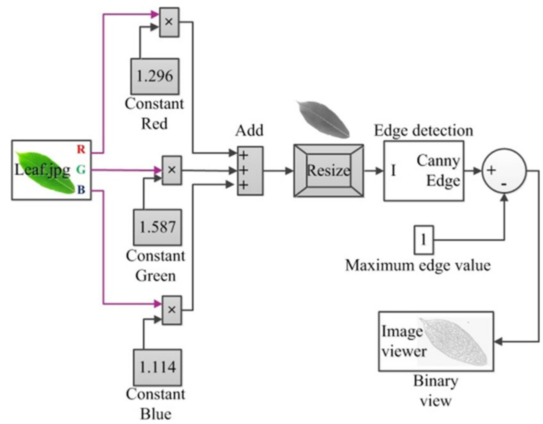


**Fig. 2** Image transformation process

### 3.1 Image acquisition and processing

Images of leaves are acquired using a handheld camera. Leaves can be categorized into two groups (i) simple and (ii) palmately compound. Images are acquired from 10 types of leaves of which 8 are simple and 2 are palmately compound types. An example of both groups of leaves is shown in Fig. 1.

A total of 600 leaves are used from different trees of the same species. Sixty leaves of each type are used. The red, green, and blue (RGB) cues of the leaf images are separated prior to converting them to a greyscale image. A scalar value of 1.2959, 1.5870, and 1.1140 is multiplied by the RGB cues. The green colour has a higher multiplier than red and blue colours cues since most leaves in nature are green. A multiplier value greater than one but less than 2 also enhances the contrast such that the edges are distinguishable. The greyscale image is resized and converted to binary form. The binary image is then inverted for feature extraction. A Simulink block diagram of image transformation from coloured to binary form is depicted in Fig. 2. The images are smoothened using a $5 \times 5$ averaging filter. Quantum edge detection shows improved precision and reduced computation complexity [27]. The colour cues

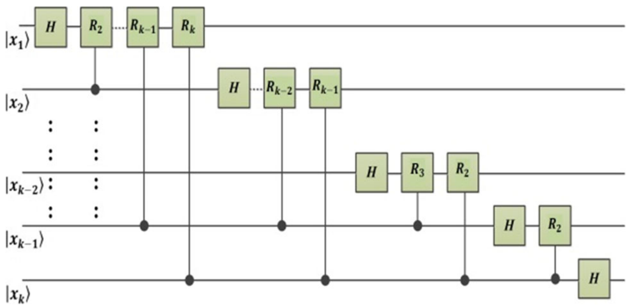


**Fig. 3** Quantum Fourier transform realisation

are entangled for data reduction. Entangling in computer vision reveals hidden features and dependency on self-occlusion (Kumar, [23].

### 3.2 Qubit representation

The images are resized to a square image ($2^n \times 2^n$) such that the number of pixels for the rows and columns are the same. The colours of the image are represented using qubits as in (1).

$$|I(k_i)\rangle = \frac{1}{2^n}\sum_{i=0}^{2^{2n}-1}\left(\cos\frac{2\pi k_i}{\aleph}\Big|0\rangle + \sin\frac{2\pi k_i}{\aleph}|1\rangle\right)\otimes|i\rangle \tag{1}$$

The size of the image is N × N (1024 × 1024), and $k_i$ is the pixel value at location $i$ and has values ranging from 0 to ℵ. Quantum Fourier transforms (QFT) are applied to the image as in (2). The QFT acts on each quantum state of the images $|I(r,c)\rangle$ defined by rows ($r$) and column ($c$) in $x$ and $y$ directions, respectively. A circuit of the realization for $n$ qubit QFT using a Hadamard gate ($H$) and rotation gate ($R_k$) is shown in Fig. 3 and described in Eqs. (3) and (4). The images are converted from pixels to states where each state is made up by averaging a square matrix. State preparation is essential in quantum machine learning for preprocessing [28]. There are different levels of state representation depending on the total pixels used in averaging. An illustration of 9 levels of image processing from qubit states is shown in Fig. 4. The approach leads to dimension reduction and preserves image information.

$$|Q(x,y)\rangle = \frac{1}{N}\sum_{r=0}^{N-1}\sum_{c=0}^{N-1}|I(r,c)\rangle\, e^{2\pi\left(\frac{x_r+y_c}{N}\right)} \tag{2}$$

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} \tag{3}$$

$$R_k = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & e^{\frac{\pi i}{2^{k-1}}}\end{bmatrix} \tag{4}$$

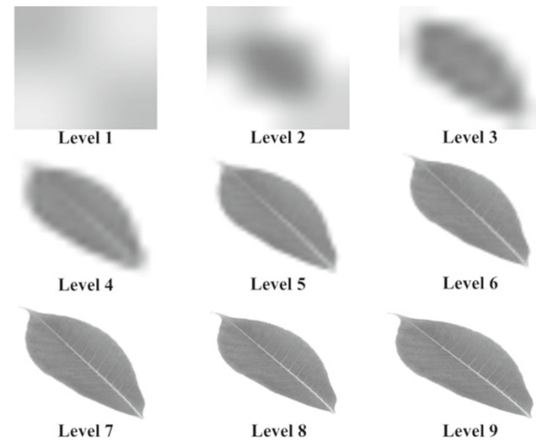


**Fig. 4** Different image processing level using qubit states

## 4 Feature extraction

Features from an image can be extracted using deep feature extraction techniques like CNN or using handcrafted geometric features [21]. The features of the leaf which are extracted are veins structure using state-based CNN, circularity, irregularity or level of belongingness to palmately, and symmetry. Features are categorized into two categories which are internal geometry and profile. The feature extraction is faster with quantum computation in comparison with classical computing. The effectiveness of quantum-based algorithm is verified in many computer vision and pattern recognition works concerning images with complex structures (Kumar, [23].

### 4.1 Internal geometry:

Vein structure and level of symmetry in leaves are regarded as internal geometric features. The features are represented as numerical vectors. The vein structures in the leaf and many other images are mostly recognized by skeletonisation techniques (Li and Liao, [26, 36], Suchitra and Gladstone, [37]. The vein features are extracted by applying a state-based CNN to the leaf image. The conventional CNN strides over the pixel, however, the proposed state-based CNN strides over the states defined by a vector. Striding uses a sliding filter with qubit gates. In [20], a 1D-convolution of the columns is done using a min–max sliding filter to evaluate the dimensions of vehicles. The pooling layer uses max-pool. The output from the pooling layer is input to a post-selection controlled by gates. A framework of state-based CNN and its quantum implementation to find end-points of veins is shown in Figs. 5 and 6, respectively. The implementation uses a counter (t), swap gates and quarter turns gate (S). A quirk simulator is used to simulate the circuit.

The symmetrical property of an image can be realised using exclusive or (XOR) gates [22]. A controlled not (CNOT) gate is used to find the symmetry in the leaf. The implementation is shown in Fig. 7, and the visuali-

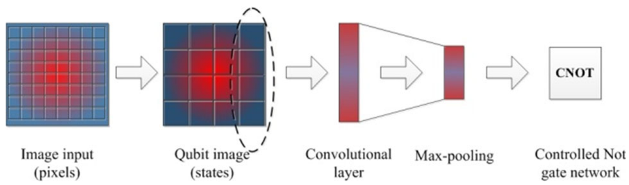


**Fig. 5** Architecture for image defined with qubits

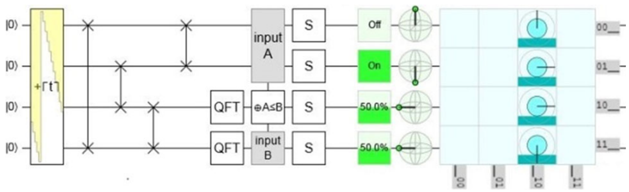

**Fig. 6** Quantum CNN with simulation realised using QFT and swap gates

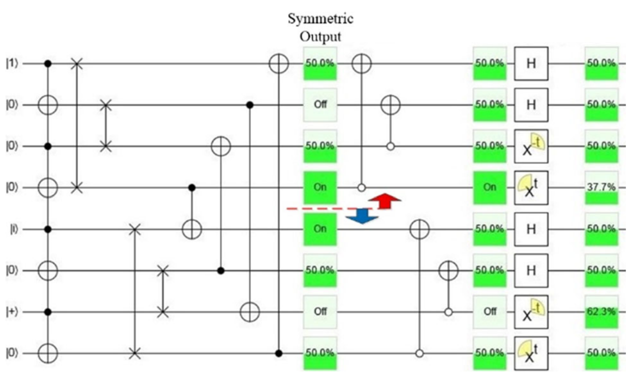


**Fig. 7** Quantum circuit for implementing the degree of symmetric properties using a combination of states shows the symmetric output

sation is shown in Figs. 8 And 9. A vertically symmetrical leaf has simulation results similar to Fig. 8. Simulation results of a complex palmately leaf and leaves with lots of curled edges are depicted in Fig. 9. Complex-shaped and curled leaves can be fragmented into zones and checked for symmetricity. The detection of curls is faster with quantum method.

The output before the CNOT gates in Figs. 7 and 8 is symmetric. The spin direction of the 4 pairs of Bloch spheres in Fig. 8 is also opposite.

## 4.2 Profile and outline

An effective way of finding circularity is by using the area to perimeter ratio. The ratio of area to perimeter of a unit circle can be used as a factor to determine circularity. The total area can be found by evaluating the sum of all states. The perimeter is evaluated by considering the total states in an edge image. The edge image is a binary image (black and white) as shown in Fig. 10.

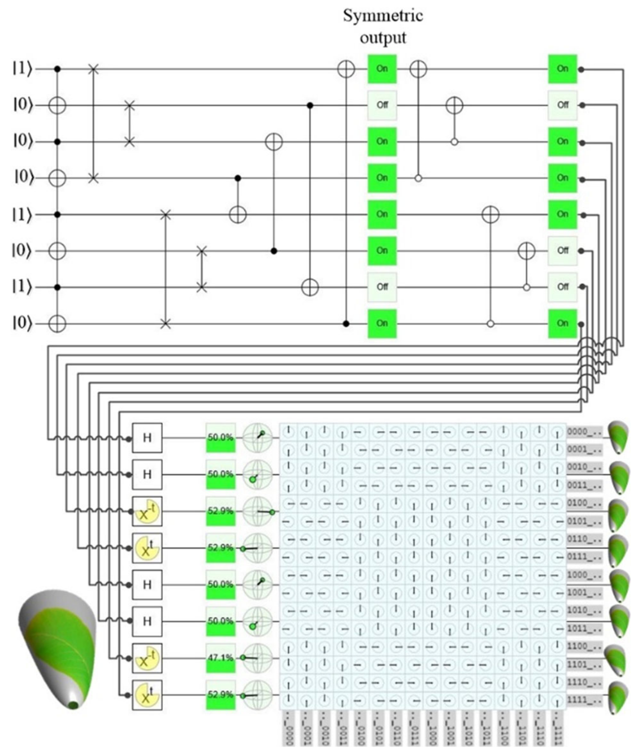


**Fig. 8** Simulated results for symmetry for an image that is vertically symmetrical

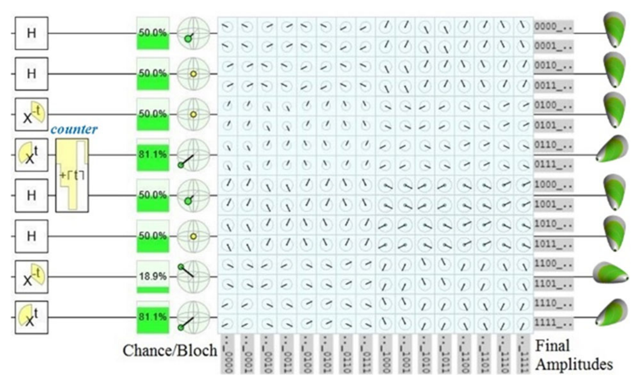


**Fig. 9** Fragmented symmetry visualization where fragments of a leaf are evaluated for symmetry

**Fig. 10** Binary edge representation of leaf

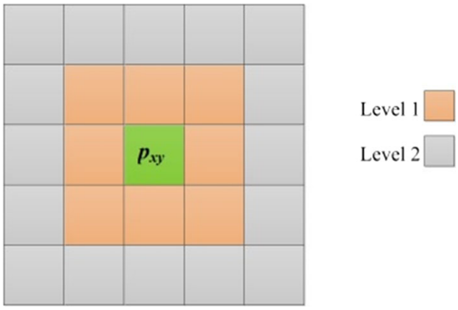


**Fig. 11** Levels of nearest neighbour for given location of qubit

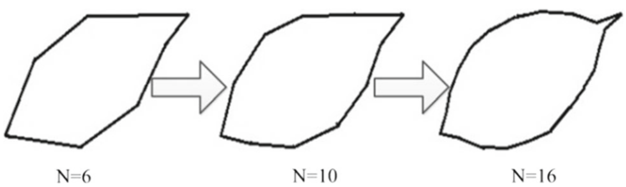


**Fig. 12** Profile of leaf based on size of samples

A more accurate method for determining the profile is by taking spatial circularity around the edge and its neighbours. The rate of change in circularity ($C_{x,y}$) at any spatial position *(x,y)* on the edges determines the profile of the leaf. The accuracy of the profile is based on the selection of samples ($N$) which divides the edge into different profiles. The first step is to check if any arbitrarily chosen qubit ($p_{x,y}$) is on the edge. The probability of a point lying on the edge is determined by the sample size and neighbouring pixels. Large subdivision of pixels will give low values of N. We use a 2-level nearest neighbour as shown in Fig. 11. Different profiles for three values of N are shown in Fig. 12.

The row-wise and column-wise convolution is done as shown in the simulations in Fig. 6. Given a point $p_{xk,yk}$, the bounds for the point are evaluated as shown;

$$(|p_{x,y} - p_{x,y_k}| \le 2) \in A, \ and \ (|p_{x,y} - p_{x_k,y}| \le 2) \in B \tag{5}$$

where $p_{x,y}$ is the nearest location of qubit bounded by the edge. We define two subgroups A and B and check if the groups match. The subgroups belong to a group $G_i$. The comparisons between subsequent groups are made to determine the gradient of the profile. The wreath product is used to match the groups. A function *S(k)* is created which compares the groups.

$$G_i = A_i \ltimes B_i \tag{6}$$

$$S(k) = G_i \wr G_{i+1} \tag{7}$$

$$S(k)_{i=1}^{N-1} \tag{8}$$

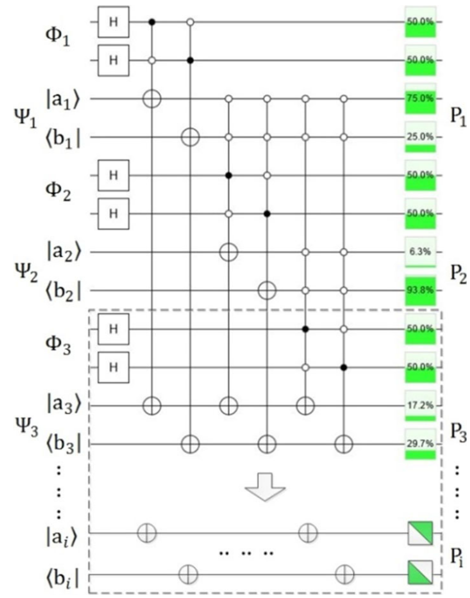


**Fig. 13** Architecture of classifier using Hadamard gates

## 5 Classification

The simplest classifiers like the nearest neighbour and decision trees can be extended by integrating quantum theories and principles. The classifiers can also be constructed using quantum gates to visualise performance.

### 5.1 Classifier

The nearest neighbour classification technique is a very simple and efficient technique used for classification. The similarity measure can also be obtained by fidelity of two quantum states and can be realised using SWAP gates [3]. The nearest neighbour technique finds the closest match based on the distance or similarity of attributes. A temporal nearest neighbour has attributes that belong to same isochronous curve and is evaluated based on shortest time. The spatial nearest neighbour considers spatial distances as measure of closeness. 65% of the images from every species and group are used for training the patterns of the leaf. The remaining 35% is used as input for testing the accuracy of the classifier. The states of the test sets ($\varPhi_i$) are matched with the train set ($S_\mathrm{i}$) having $n$ labels.

$$|\varnothing_i\rangle : i \in \{0, \cdots, n-1\} \tag{9}$$

$$S = [S_0, \cdots, S_{n-1}] \tag{10}$$

The states are super-imposed, and the probability is evaluated for profile-based features and for structural features. A circuit implementation with simulation for three classes is shown in Fig. 13. The figure predicts the belongingness of a test vector to a class or species based on patterns from feature extraction. It also gives the probability of the leaf's belongingness to a group (palmately or simple). The conventional nearest neighbour classification lacks the probability assignment of a test object to a class label. The probability is based on a square tiling matrix $M$. The test vector is divided into two states with equal probability, $\Psi = [\mathrm{P_k}\ 1\text{-}\mathrm{P_k}]$. The two states of the test vector are internal geometry features and profile features. Both the features are equally weighted and thus allocated 50% input towards correct classification.

$$M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{11}$$

A Hadamard gate is used with controlled not gates to evaluate the probabilities. The Hadamard gate for $i$ number of states is given in (12). The probability is calculated by multiplying each state of the test vector with a controlled Hadamard gate. Figure 13 shows that each state control one Hadamard gate.

$$H_i = \frac{1}{\sqrt{2}^i} \begin{bmatrix} M_{1,1} & M_{1,2} & \dots & M_{1,i} \\ M_{2,1} & M_{2,2} & \dots & M_{2,i} \\ \vdots & \vdots & \ddots & \vdots \\ M_{i,1} & M_{i,2} & \dots & M_{i,i} \end{bmatrix} \tag{12}$$

$$P = H[\Psi_1, \cdots, \Psi_i]^T \tag{13}$$

### 5.2 Results

The accuracy of classification is determined for the entire test set by evaluating the total of correctly classified leaves. The accuracy of the quantum-based approach is compared with the classical nearest neighbour. The feature vectors of the conventional nearest neighbour classifier differ from the quantum states feature vectors. The quantum version of images is defined using qubits, unlike the original images which are defined by pixels and colour cues. The results are shown in Table 1. The confidence error of accuracy for each type of leaf and each group of leaves are also presented in Table 1. The error is based on a 95% confidence interval.

The accuracy using different levels of image processing is also tested using the nearest neighbour classifier. The test samples are kept same for different levels of image processing. The feature extraction techniques are also kept the same. A graph of accuracy based on different levels of image processing is shown in Fig. 14. The results show that the accuracy of classification is highest with 9 and 10 levels of image processing. The original size of images is 1024 × 1024 ($2^{10} \times 2^{10}$), while the $9^{\text{th}}$ level of image processing on qubits is 512 × 512 pixels or 9 × 9 qubit states. There is a huge dimensionality reduction in images. The informative features of images are preserved yielding around 90% accuracy (see Fig. 14).

The proposed method is compared with other works in leaf recognition (see Table 2). The comparison is made in terms of recognition rate. An average of 88.78%

**Table 1** Recognition rate using quantum nearest neighbour and classical nearest neighbour classifier

| Labels | Leaf type | Accuracy (classical) | Accuracy (quantum) (%) |
|---|---|---|---|
| 1 | Simple | 92.83 ± 0.10% | 86.67 ± 0.15 |
| 2 | Simple | 85.67 ± 0.21% | 92.83 ± 0.08 |
| 3 | Simple | 84.67 ± 0.22% | 89.00 ± 0.13 |
| 4 | Simple | 66.67 ± 0.48% | 68.33 ± 0.35 |
| 5 | Simple | 66.67 ± 0.48% | 74.50 ± 0.28 |
| 6 | Simple | 80.00 ± 0.29% | 89.67 ± 0.12 |
| 7 | Simple | 53.83 ± 0.66% | 65.33 ± 0.39 |
| 8 | Simple | 73.33 ± 0.38% | 75.50 ± 0.27 |
| 9 | Palmately | 65.00 ± 0.50% | 94.66 ± 0.06 |
| 10 | Palmately | 71.67 ± 0.41% | 100 ± 0 |
| Overall | Simple | 75.46 ± 0.35% | 80.23 ± 0.26 |
| | Palmately | 68.33 ± 0.48% | 97.33 ± 0.01 |

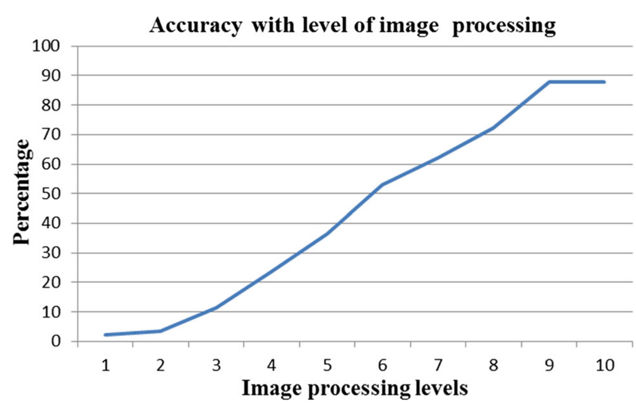


**Fig. 14** Accuracy in classification for different input images based on the level of image processing

**Table 2** Comparison of proposed quantum classifier result with previous related research

| Author | Method | Accuracy (%) |
|---|---|---|
| [39] | Support vector machine (SVM) | 97.8 |
| [10] | Hierarchical descriptors | 95.4 |
| [4] | DHGN + k-NN | 71.4 |
| [44] | k -SVNN | 71.3 |
| [40] | PNN | 90.0 |
| Proposed | Quantum classifier | 97.3 |

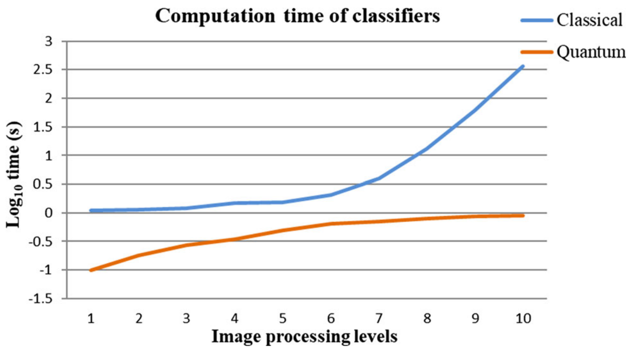


**Fig. 15** Logarithmic computing time (in seconds) for classical and quantum classifiers for 10 level of image processing

accuracy is achieved for overall leaf recognition. The quantum classifier is able to classify complex palmately leaves with 97.33% accuracy. Simple leaves were classified with 80.23% accuracy. The proposed quantum classifier has achieved good accuracy compared to similar works of others. The average accuracy of the proposed system is 88.78%. The classical model has a 25.97% error, whereas the quantum version of the classifier has a 16.35% of model error.

The quantum classifier used is based on probability and can tell the probability of belongingness of a species of leaf to a class label. The quantum nearest neighbour classification has fairly good performance for simple type of leaves. Comparing the two methods of feature extraction, the 1D convolutional with QFT can detect complex leaves with more robustness. The profile features are more robust in classifying simple leaves.

The performance of the classification is also evaluated for computation complexity. The computation time includes the time taken for image conversion and processing, feature extraction, and classification. The classical model does not have a conversion process. The total computation time is evaluated for all the levels of image processing. The performance of the computation time is shown in Fig. 15. The number of quantum gates employed in the quantum circuit determine the computation complexity of classification. Quantum classifier has a low computation time in comparison with the conventional classic method of classification. The computation time for image processing and classification using the classical classifier increases exponentially with increase in image resolution.

## 6 Conclusion

A quantum classifier that uses nearest neighbour searching is developed to classify between 10 classes of leaves. The proposed classifier outperforms the conventional nearest neighbour classification technique. The quantum classifier also shows the probability of belongingness of a test leaf to different species thus reducing computation time. Results show that our classification is 88.78% accurate and can classify complex-shaped leaves (palmately type leaves) with 97.33% accuracy. The proposed method can be applied to other types of leaves and image classification systems.

Due to the complex nature of quantum mechanics, the work has been limited to adapting simple classifiers. The quantum version of the simplest nearest neighbour classifier has better accuracy than the classical nearest neighbour classifier. It is anticipated that the quantum version of deep learning models will also have a better performance than the simplest deep learning model. Future work is to extend the quantum classifier to a multi-view computer vision network, use other aspects like leaf shapes for recognition, and integrate quantum mechanics in the existing deep learning models of computer vision systems.

## Author contributions

All authors have contributed to the manuscript. The contribution includes preparing materials, data collection, analysis, simulation, and reviewing of the manuscript. All the authors approved the final manuscript.

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**Data Availability Statement** This manuscript has no associated data or the data will not be deposited. The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

### Declarations

**Conflict of interest** The authors declare no conflict of interest and declare that this manuscript is original work.

**Human and animal rights** The research did not involve any animals or human participants.

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